

SHARANYA AHIRE

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EDUCATION

Vellore Institute of Technology, Chennai B.Tech Computer Science and Engineering (AI & ML) CGPA: 8.62	Jul 2024 – Present
New Bombay City School, Ghansoli (CBSE) Class XII Percentage: 83%	2022 – 2024
New Horizon Public School (CBSE) Class X Percentage: 92%	2020 – 2022

TECHNICAL SKILLS

- **Languages:** Python, C, C++, SQL, JavaScript
- **Frontend:** React.js, HTML, CSS, Tailwind CSS
- **Backend:** Node.js, Express.js
- **AI/ML:** PyTorch, OpenCV, NumPy, Pillow, Scikit-image, MediaPipe
- **Tools:** Git, GitHub, VS Code, Google Colab, Vercel, Render
- **Concepts:** Object-Oriented Programming, Data Structures & Algorithms, Artificial Intelligence, Machine Learning, Computer Vision

PROJECTS

Sahayata – AI Powered Legal Assistance Platform

- Developed a full-stack AI-powered legal assistance platform to provide accessible legal guidance and emergency support.
- Integrated GPT-powered legal assistance with multilingual voice interaction for simplified legal awareness.
- Implemented location-based police station discovery using Google Maps API and a verified lawyer directory.
- Built using React.js, Node.js, Express.js, Tailwind CSS and deployed using Vercel and Render.

Intelligent Emergency Rescue Planner

- Developed an AI-based emergency response planning system for optimal ambulance allocation and rescue routing.
- Implemented A*, Breadth First Search (BFS) and Uniform Cost Search (UCS) algorithms for intelligent path planning.
- Designed a rescue planner to allocate the most suitable ambulance based on emergency location and estimated road conditions.
- Integrated Bayesian probability estimation, graph visualization and ambulance simulation for real-time rescue planning.

Hand Gesture Virtual Drawing using Computer Vision

- Developed a virtual drawing application using MediaPipe hand tracking for real-time gesture recognition.
- Implemented smooth freehand drawing capable of generating continuous curves using fingertip trajectory tracking.
- Designed gesture controls for brush selection, eraser mode, dynamic brush size adjustment and canvas clearing.
- Built using React.js, JavaScript, HTML and CSS.

RESEARCH EXPERIENCE

Summer Research Internship Program (SRIP)

Vellore Institute of Technology

- Conducted research on deep learning-based restoration of degraded chest X-ray images for medical imaging applications.
- Implemented and evaluated state-of-the-art image restoration architectures including **DnCNN**, **SwinIR**, **Restormer** and **NAFNet**.

- Built preprocessing pipelines for generating degraded medical datasets and trained supervised restoration models using PyTorch.
- Performed comparative analysis of restoration models through extensive experimentation and documentation.

LEADERSHIP & RESPONSIBILITIES

Vice Chairperson

Swarajya – Marathi Literary Association

- Led planning and execution of cultural and literary events.
- Coordinated volunteer teams, event logistics and student engagement initiatives.

Volunteer

Technovit 2025 – Special Teams Committee

- Assisted in organizing and managing technical fest operations and event coordination.

Volunteer

Vibrance 2025 – Special Guest Care Committee

- Managed hospitality, logistics and guest coordination during the university cultural festival.

CERTIFICATIONS

- IBM Agentic AI Professional Certificate
- NVIDIA Fundamentals of Deep Learning
- NVIDIA Fundamentals of Accelerated Computing with CUDA C/C++